# Colouring a Poisson–Dirichlet process recovers the Dirichlet distribution: a probabilistic proof

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doi:10.5802/jep.317; CC BY 4.0

*Editorial note.* The counted statement is the author's labelled proposition 'prop:probabilistic dirichlet law' (source lines 1121-1128); the proof environment begins on the next line after \end{prop} apart from one blank line, so the pairing is by adjacency.

*Editorial note.* The retained context is the appendix passage that contains the colouring theorem and Campbell's theorem, both of which the retained proof invokes by label; retaining it keeps every used label defined inside the package.

*Editorial note.* The citations inside the retained text and context resolve against the publisher bibliography retained with this source.

*Editorial note.* No counter adjustment is retained: the publisher numbering is not reproduced and the item prints with the wrapper's own sequential numbering.

*Editorial scope.* One original result of the pinned publisher article, counted once: the probabilistic derivation of the Dirichlet law by colouring a Poisson–Dirichlet process (author label prop:probabilistic dirichlet law).

*Allowed standard prerequisites (editorial).*

- Poisson–Dirichlet process. The normalised points V_i = X_i/S of a Poisson process on the positive half-line with intensity e^{-x}/x dx form a Poisson–Dirichlet process.. It is the representation adopted at source line 1131.
- Colouring theorem. Randomly colouring the points of a Poisson process with k colours yields independent Poisson processes with intensities p_i times the original intensity.. It is applied at source line 1131 to split the process into the k coloured subprocesses Pi_m.
- Campbell's theorem. For a Poisson process, the Laplace transform of the sum of a functional over the points is the exponential of the corresponding integral.. It computes the moment generating function of S_m at source lines 1139-1142.

**Proposition 0.1** (Coloring theorem, [11, §5.1])**.** *If* $\Pi$ *is a Poisson process with mean measure* $\mu$ *on* $S$*, and the points are randomly colored with* $k$ *colors such that the probability of a point receiving the color* $i$ *is* $p_i$*, and such that the color*

*of a point is independent of different points and of the position of the point. Let* $\Pi_i$ *be the subset of* $\Pi$ *with color* $i$*. Then all the* $\Pi_i$ *are independent Poisson processes with mean measures* $\mu_i = p_i\mu$*.*

**Proposition 0.2** (Campbell's theorem, [11, §3.2])**.** *Let* $\Pi$ *be any Poisson process on* $S$ *with mean measure* $\mu$*. Let* $f\colon S \to \mathbb{R}$ *be a measurable function. Then*

$$\Sigma = \sum_{X\in\Pi} f(X)$$

*is absolutely convergent almost surely if, and only if,*

$$\int_S \min\{f, 1\}\,\mathrm{d}\mu < \infty.$$

*If this condition holds, then*

$$\mathbb{E}[e^{s\Sigma}] = \exp\left(\int_S (e^{sf} - 1)\,\mathrm{d}\mu\right)$$

*for any complex* $s$ *for which the integral converges. Moreover,*

$$\mathbb{E}[\Sigma] = \int_S f\,\mathrm{d}\mu$$

*if the integral converges. In the case where it converges, we also have*

$$\mathrm{Var}[\Sigma] = \int_S f^2\,\mathrm{d}\mu.$$

**Proposition 0.3** (Donnelly–Tavaré [8])**.** *Let* $\boldsymbol{V} = (V_1, V_2, \ldots)$ *be a Poisson–Dirichlet process, let* $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_k) \in \Delta^{k-1}$ *such that* $\alpha_i > 0$ *for all* $1 \le i \le k$*, and let* $(C_i)_{i\ge1}$ *be a sequence of i.i.d. random variables (also independent of* $\boldsymbol{V}$*) with* $\mathbb{P}[C_i = j] = \alpha_j$ *for all* $i \ge 1$*. Then the random vector*

$$\Big(\sum_{i:\ C_i=1} V_i, \ldots, \sum_{i:\ C_i=k} V_i\Big)$$

*is distributed according to* $\mathrm{Dir}(\boldsymbol{\alpha})$*.*

*Proof.* Let $X_1 > X_2 > \cdots$ be the points of a Poisson process on $\mathbb{R}_{>0}$ with intensity $(e^{-x}/x)\,\mathrm{d}x$, and let $S \coloneqq \sum_{i\ge1} X_i$. For each $i \ge 1$, Let $V_i \coloneqq X_i/S$. We then know that $(V_1, V_2, \ldots)$ follows the Poisson–Dirichlet distribution [9, Th. 2.2]. With the coloring theorem (Proposition 0.1), the point processes

$$\Pi_m = \{X_i : C_i = m\}$$

form independent Poisson processes of intensity $\alpha_m \cdot (e^{-x}/x)\,\mathrm{d}x$ for all $m = 1, \ldots, k$. We then let

$$S_m \coloneqq \sum_{X\in\Pi_m} X$$

for $m = 1, 2, \ldots, k$. With Campbell's theorem (Proposition 0.2), we compute the moment generating function

$$\log \mathbb{E}[e^{sS_m}] = \alpha_m \cdot \int_0^\infty \frac{e^{sx} - 1}{x} \cdot e^{-x}\,\mathrm{d}x = -\alpha_m \log(1 - s)$$

for $\mathrm{Re}(s) < 1$. This coincides with the moment generating function of Gamma$(\alpha_m, 1)$ distribution. We thus deduce that the vector

$$\Big( \sum_{i:\ C_i=1} V_i, \ldots, \sum_{i:\ C_i=k} V_i \Big) = \Big( \frac{S_1}{\sum_{m=1}^k S_m}, \ldots, \frac{S_k}{\sum_{m=1}^k S_m} \Big)$$

follows the distribution Dir$(\boldsymbol{\alpha})$ (see [12, Ch. 49, pp. 485-487] for a proof). □

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